

1 Project Overview

SafeSite AI is an intelligent construction safety monitoring platform that utilizes Computer Vision and Edge Artificial Intelligence to automatically detect unsafe worker behavior from construction site video feeds.

The system continuously monitors worker activities using CCTV cameras and identifies safety violations such as missing Personal Protective Equipment (PPE), unsafe posture, restricted zone intrusion, and dangerous proximity to heavy machinery.

Unlike traditional monitoring systems, SafeSite AI performs real-time risk assessment and generates actionable alerts before accidents occur.

2 Problem Statement

Construction sites remain among the most hazardous workplaces worldwide. Accidents frequently occur because of:

- Failure to wear safety helmets and reflective vests.
- Unsafe worker behavior and posture.
- Restricted zone violations.
- Dangerous interaction with heavy machinery.
- Lack of continuous supervision.

Current monitoring approaches depend heavily on manual observation, making continuous safety enforcement difficult and prone to human error.

3 Proposed Solution

SafeSite AI processes live CCTV video feeds using Edge AI devices deployed directly at construction sites.

The platform automatically detects:

- Missing Safety Helmets
- Missing Safety Vests
- Unsafe Worker Posture
- Hazard Zone Intrusion
- Worker-Machinery Proximity Risks
- High-Risk Activities

Whenever a violation is detected, the system generates real-time alerts and updates a centralized monitoring dashboard.

4 Core Features

PPE Compliance Detection

The system detects helmets and safety vests and identifies workers violating safety regulations.

Restricted Zone Monitoring

Virtual safety boundaries are established around hazardous areas such as crane operating zones and excavation sites.

Unsafe Behavior Detection

Using Human Pose Estimation, the system identifies unsafe climbing, dangerous posture, improper lifting, and fall-risk situations.

Worker-Machinery Proximity Detection

The platform continuously monitors worker distance from cranes, excavators, forklifts, and construction vehicles.

Context-Aware Risk Engine

The system combines multiple violations to generate dynamic risk scores and prioritize critical hazards.

5 Technology & Implementation

Artificial Intelligence

YOLO Object Detection, OpenCV, MediaPipe, TensorFlow Lite

Backend

Node.js, Express.js, REST APIs

Database

MongoDB

Frontend

Next.js, Tailwind CSS

Edge Hardware

Raspberry Pi / NVIDIA Jetson Nano

Implementation Methodology

1. Detect workers, helmets, and safety vests using YOLO.
2. Monitor worker movement inside restricted zones.
3. Detect machinery and calculate worker-machinery proximity.
4. Apply Human Pose Estimation for unsafe behavior detection.
5. Generate contextual risk scores using detected violations.
6. Display alerts and analytics on a centralized dashboard.

Risk Assessment Framework

Violation	Risk Score
Helmet Missing	20
Safety Vest Missing	15
Restricted Zone Entry	30
Machinery Proximity	25
Unsafe Posture	10

Risk Levels:

- 0–25 : Low Risk
- 26–50 : Medium Risk

- 51–75 : High Risk
- Above 75 : Critical Risk

6 Real-World Deployment and Working

SafeSite AI is designed to operate directly on active construction sites using existing CCTV infrastructure and Edge AI devices.

Step 1: Camera Placement

High-resolution CCTV cameras will be installed at strategic locations across the construction site, including:

- Site Entry and Exit Gates
- Crane Operating Areas
- Excavation Zones
- Material Handling Areas
- Elevated Working Platforms
- Heavy Machinery Operating Regions

These locations are selected because they represent the highest-risk areas for workplace accidents.

Step 2: Video Capture

The CCTV cameras continuously capture video streams of worker activities and site operations. Instead of sending raw video directly to cloud servers, the video feed is forwarded to a nearby Edge AI device.

Step 3: Edge AI Processing

A Raspberry Pi or NVIDIA Jetson Nano deployed inside the site office processes the video locally. The Edge AI device performs:

- Worker Detection
- Helmet Detection
- Safety Vest Detection
- Machinery Detection
- Human Pose Estimation
- Restricted Zone Monitoring

Local processing reduces network dependency and enables real-time decision making.

Step 4: Violation Detection

The AI system continuously analyzes worker activities and identifies safety violations such as:

- Missing Safety Helmet
- Missing Safety Vest
- Entry into Restricted Areas

- Unsafe Worker Posture
- Unsafe Proximity to Heavy Machinery

Detected violations are immediately forwarded to the Risk Assessment Engine.

Step 5: Risk Assessment

The Risk Engine combines multiple detected violations to calculate an overall risk score. For example:

Worker without Helmet
+ Standing inside Crane Operating Zone
+ Working close to Excavator
= Critical Risk

This helps prioritize the most dangerous situations.

Step 6: Alert Generation

Whenever the risk score exceeds predefined thresholds, alerts are generated automatically. Alerts may include:

- Dashboard Notifications
- SMS Alerts
- Email Notifications
- Supervisor Alerts

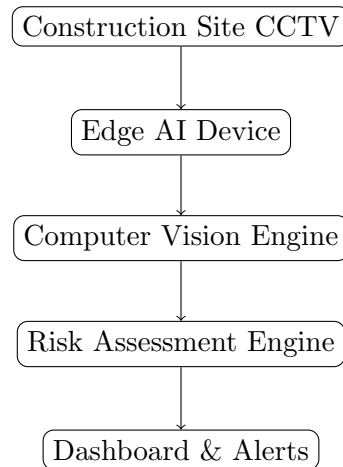
Step 7: Monitoring Dashboard

Safety officers access a centralized dashboard displaying:

- Live Camera Feeds
- Active Workers
- Current Violations
- Risk Levels
- Compliance Statistics
- Historical Safety Reports

The dashboard enables continuous site-wide safety monitoring and rapid incident response.

7 System Architecture



8 Development Roadmap

1. Requirement Analysis and Research
2. Dataset Collection and Annotation
3. PPE Detection Development
4. Unsafe Behavior Detection
5. Risk Assessment Engine
6. Dashboard Development
7. Edge Deployment
8. Testing and Optimization
9. Final Demonstration

9 Expected Outcomes

The project aims to deliver:

- Automated Safety Monitoring
- Reduced Workplace Accidents
- Improved PPE Compliance
- Real-Time Hazard Detection
- Faster Incident Response
- Intelligent Safety Analytics

Applications

Construction Sites, Industrial Plants, Manufacturing Facilities, Warehouses, Mining Operations, and Infrastructure Development Projects.

10 Conclusion

SafeSite AI presents a practical and scalable solution for improving construction worker safety through Artificial Intelligence and Edge Computing.

The proposed system enables continuous monitoring, intelligent risk assessment, and proactive accident prevention. By identifying safety violations before incidents occur, SafeSite AI contributes toward safer, smarter, and more efficient construction environments.

Future enhancements may include drone-based monitoring, worker fatigue detection, multi-camera tracking, and predictive accident forecasting.